

490 Final Presentation

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- **W**e are the lead of a group of virologists in the face of a new disease.
- **M**oving forward efficiently is key.
- **I**llness leaves patients weak, they can only handle so many drugs.
- **H**ow do we balance speed and ensuring our cure will work?

- **C**omparing the effectiveness of two or more options is done through ANOVA.
- **A**NOVA is only the beginning of our process.
- **P**ost hoc tests tell us what ANOVA cannot.
- **W**hat post hoc test should we use?

Post Hoc Tests

- The selected tests represent a good variety of attributes (power, liberal vs conservative, specialty).
- Bonferroni Correction
- Tukey's HSD
- Scheffe's Method
- Fisher's LSD
- Holm-Bonferroni
- Dunnett's Test
- Benjamini-Hochberg (BH)

Data Generation Process

- **U**se the `rnorm` function to generate our data.
- **W**rite a function to simulate these 1000 times each.
- **R**un ANOVA on each triad.
- **R**un each post hoc test on every triad and examine their output.

Assumptions

- **N**ormality
- **I**ndependence
- **H**omoscedasticity

Test Results

	Prop Significant	Average P-value
Old vs New	0.394	2.631323e-01
Placebo vs New	1.000	6.733524e-05
Placebo vs Old	0.968	7.271139e-03

Table: Bonferroni Correction Summary Table

	Prop Significant	Average P-value
Old vs New	0.433	1.958669e-01
Placebo vs New	1.000	6.273578e-05
Placebo vs Old	0.974	5.958850e-03

Table: Tukey's HSD Summary Table

Test Results

	Prop Significant	Average P-value
Old vs New	0.377	0.2112300948
Placebo vs New	1.000	0.0000931723
Placebo vs Old	0.960	0.0074852060

Table: Scheffe's Method Summary Table

	Prop Significant	Average P-value
Old vs New	0.595	1.143169e-01
Placebo vs New	1.000	2.244508e-05
Placebo vs Old	0.991	2.423713e-03

Table: Fisher's LSD Summary Table

Test Results

	Prop Significant	Average P-value
Old vs New	NA	NA
Placebo vs New	1.000	4.397824e-05
Placebo vs Old	0.985	4.516697e-03

Table: Dunnett's Test Summary Table

*Old vs New not ran to simulate the limitations of this test.

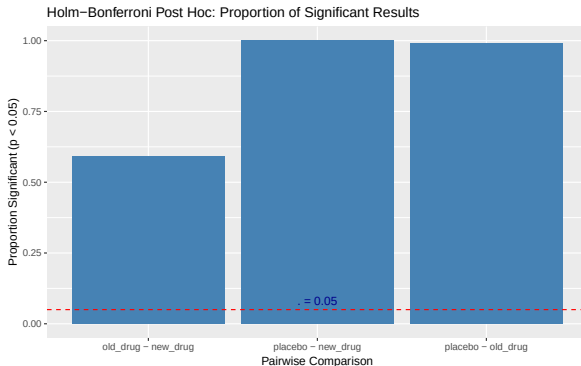
	Prop Significant	Average P-value
Old vs New	0.594	1.144483e-01
Placebo vs New	1.000	6.233299e-05
Placebo vs Old	0.991	2.791599e-03

Table: Benjamini-Hochberg Summary Table

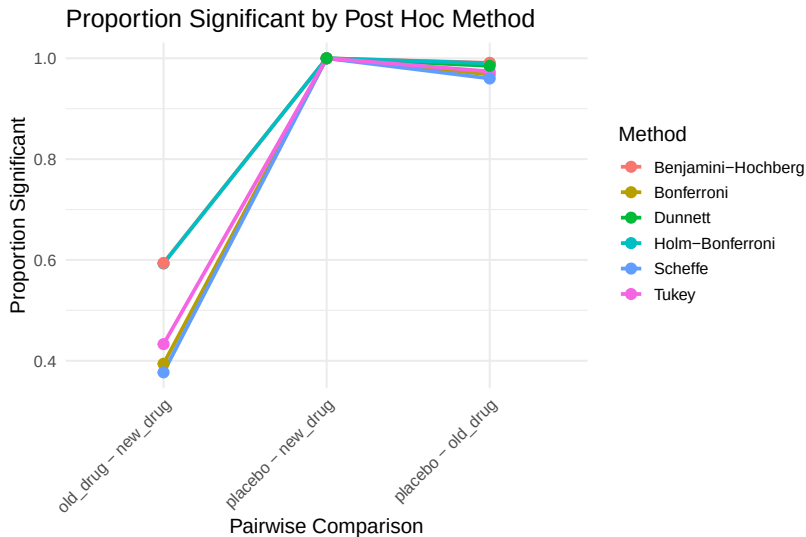
Holm-Bonferroni Results

	Prop Significant	Average P-value
Old vs New	0.593	1.146882e-01
Placebo vs New	1.000	6.417125e-05
Placebo vs Old	0.990	3.264599e-03

Table: Holm-Bonferroni Summary Table



Comparison Graph



Conclusion

- **W**hy is Holm-Bonferroni the best?
- **W**hat does it mean for it to be conservative?
- **I**f it's conservative, how does it perform as well as Fisher's LSD?
- **I**f Holm-Bonferroni is the best, why do the other tests matter?

References

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Thank you for your time

Questions?